■# PAPER_308 — Spiral Arm Torque Gravitational Amplifier

Session 88 | 30th C++ UQFF module | FIRST Spiral+SN Ia UQFF 2.0

■# PAPER_308 — Spiral Arm Torque Gravitational Amplifier

τspiral(10 Gyr) = 2.046 | Tpattern = 307 Myr | dτ/dt = 2.741 × H■_SH0ES

Session 88 | 30th C++ UQFF module | FIRST Spiral+SN Ia UQFF 2.0 **Module:** SPIRAL*SUPERNOVAE*UQFF_MODULE.cpp **Classification:** FIRST UQFF spiral arm pattern speed gravitational amplifier **Status:** Unique physics — no prior UQFF spiral torque gravity amplification study

---

Abstract

Spiral galaxies evolve under pattern-speed-driven star-formation torques whose cumulative gravitational amplitudes differ fundamentally from static NFW or Keplerian assumptions. Within the UQFF (Unified Quantum Field Framework) 2.0 pipeline, a dimensionless spiral torque factor τ*spiral = (M*gas/M) × Ω*p × t accumulates over cosmic time and multiplies the base gravitational term. At 10 Gyr the amplifier reaches τ = 2.046, boosting effective gravity by a factor of 3.046×. The torque rate dτ/dt = 6.483 × 10^{-1}■ s^{-1} exceeds H*■SH0ES (73.0 km/s/Mpc = 2.366 × 10^{-1}■ s^{-1}) by a factor of 2.741, establishing a new UQFF cosmic-rate comparison.

---

2. System Parameters

Parameter	Value	Notes
M (galaxy)	1.989 × 10■^{1} kg	1 × 10^{11} M_sun (Milky-Way class)
M_gas (arm gas)	1.989 × 10^{3}■ kg	1 × 10■ M_sun
r	9.258 × 10^{20} m	~30 kpc half-radius
Ω_p (pattern speed)	6.483 × 10^{-1}■ rad/s	20 km/s/kpc
H■_SH0ES	73.0 km/s/Mpc	SH0ES, Riess et al. 2022
f*gas* = *M*gas/M	0.01	Gas arm mass fraction

---

3. Derivation

3.1 Dimensionless Spiral Torque

The spiral arm torque amplifier in UQFF 2.0 is defined as a dimensionless running accumulation of pattern momentum:

```
\tau_{\rm spiral}(t) = \frac{M_{\rm gas}}{M} \cdot \Omega_p \cdot t
```

f_gas = 10■ / 10^{11} = **0.01**

Ω_p = 20.0 × 10^{3} / 3.086 × 10^{1}■ = **6.483 × 10^{-1}■ rad/s**

t (10 Gyr) = 10 × 10■ × 3.15576 × 10■ = **3.156 × 10^{1}■ s**

$$\tau_{\rm spiral}(10\,{\rm Gyr}) = 0.01 \times 6.483\times10^{-16} \times 3.156\times10^{17} = \boxed{2.046}$$

3.2 Gravity Amplification

The UQFF 2.0 pipeline applies τ as a multiplicative stage:

$$g_{\rm pipeline}(t) = \frac{GM}{r^2} \cdot \underbrace{(1 + H_z\,t)}_{\text{Hubble}} \cdot \underbrace{(1 + \tau_{\rm spiral})}_{\text{torque}} \cdot \underbrace{(1 - B/B_{\rm crit})}_{\text{SC}} \cdot \underbrace{(1 + f_{\rm TRZ})}_{\rm TRZ}$$

At t = 10 Gyr, z = 0:

$$g_{\rm amp} = 1 + \tau_{\rm spiral} = 1 + 2.046 = \boxed{3.046}$$

Effective gravity is **3× stronger at 10 Gyr** than at formation, driven entirely by spiral arm pattern momentum accumulation.

3.3 Pattern Period

$$T_{\rm pattern} = \frac{2\pi}{\Omega_p} = \frac{2\pi}{6.483\times10^{-16}} = 9.692\times10^{15}\,{\rm s} = \boxed{307 \text{ Myr}}$$

A spiral arm completes one full rotation pattern in 307 Myr — consistent with observed grand-design spiral arm lifetimes.

3.4 Torque Rate vs Hubble Rate

$$\frac{d\tau}{dt} = f_{\rm gas} \cdot \Omega_p = 6.483\times10^{-18}\,{\rm s}^{-1}$$

$$H_{0,\rm SH0ES} = \frac{73.0 \times 10^3}{3.086\times10^{22}} = 2.366\times10^{-18}\,{\rm s}^{-1}$$

$$\frac{d\tau/dt}{H_0} = \frac{6.483\times10^{-18}}{2.366\times10^{-18}} = \boxed{2.741}$$

The spiral torque accumulation rate exceeds the Hubble expansion rate by 2.741×. This means spiral arm gravitational evolution is cosmologically faster than universal expansion, a key UQFF prediction distinguishing galactic internal dynamics from global metric expansion.

---

4. Physical Interpretation

The spiral arm torque factor captures how non-axisymmetric mass flows (gas infall, density wave sweeping, star formation) cumulatively torque the gravitational potential. In UQFF 2.0, this is written as a dimensionless running factor on the total gravity pipeline. The key results:

- **τ_spiral(10 Gyr) = 2.046** — Spiral galaxies experience >3× internal gravity enhancement relative to their formation epoch
- **T_pattern = 307 Myr** — Sets the natural clock for spiral arm-driven enrichment cycles (OB association lifetimes, molecular cloud compression cycles)
- **dτ/dt / H■ = 2.741** — Torque evolution rate 2.7× faster than cosmic expansion: internal galactic structure evolves more rapidly than expanding-universe metrics suggest

---

5. UQFF 2.0 Equations (Full Pipeline, t = 10 Gyr)

Term	Value	Notes
g_base = GM/r^2	1.549×10^{-11} m/s^2	Reference gravity at 30 kpc
Hubble factor (1 + Hz·t)	system-z dependent	Expansion correction
Torque factor (1 + τ)	**3.046**	PAPER_308 key result at 10 Gyr
SC factor (1 – B/B_crit)	≈ 1.0	Galactic B ■ B_crit
TRZ factor (1 + f_TRZ)	1.10	Resonance zone correction
Ug_sum	2 × g_base	Magnetic-dipole + vacuum-SC
Λ term	~ 0	Sub-dominant at galactic scales
Quantum term	■/(m_H·Δx^2)	HUP contribution
g_DM	1.316×10^{-11} m/s^2	PAPER_310 dark matter partition
a_SN	3.096 × 10■ m/s^2	PAPER_309 SN Ia radiation pressure

6. Novel Findings (UQFF Firsts)

FIRST UQFF spiral arm pattern speed gravitational amplifier

FIRST UQFF dimensionless torque running factor in 9-term pipeline

FIRST UQFF galactic-scale dτ/dt vs H■ comparative rate analysis

τ = 2.046 exceeds unity: spiral arm evolution **non-perturbatively** amplifies gravity at late cosmic time

7. References

Riess et al. 2022, ApJ 934 L7 (SH0ES H■ = 73.04 km/s/Mpc)

Binney & Tremaine, *Galactic Dynamics* 2nd ed. (spiral arm pattern speeds)

UQFF 2.0 Architecture — ARCHITECTURE*FLOW*DIAGRAM.md v4.4.0 CANONICAL

MAIN*1*CoAnQi.cpp SOURCE4 (lines 25623–26026) — UQFF/MUGE framework

Session 88 — SPIRAL*SUPERNOVAE*UQFF_MODULE.cpp (UQFF 2.0 implementation)